

WASP-11b

R-2188

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_101012 · created 2026-08-13T23:33:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455481.848221 +/- 1.4e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.065 +/- 0.05
Transit depth (Rp/Rs)^2	0.42 +/- 0.66 [%]
Orbital Inclination (inc)	84.23 +/- 1.72
Airmass coefficient 1 (a1)	0.14 +/- 0.13
Airmass coefficient 2 (a2)	2.08 +/- 1.58
Scatter in the residuals of the lightcurve fit is	113.61 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	3.25
Transit Duration (day)	0.021 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.065 ± 0.05 vs 0.131 (deviation 1.32σ)
Duration fit vs published	0.021 d vs 0.1075 d (deviation 2.79σ)
Mid-time O–C	-0.2 min
Depth SNR	1.3
Residual scatter	113.61 %

Light curve

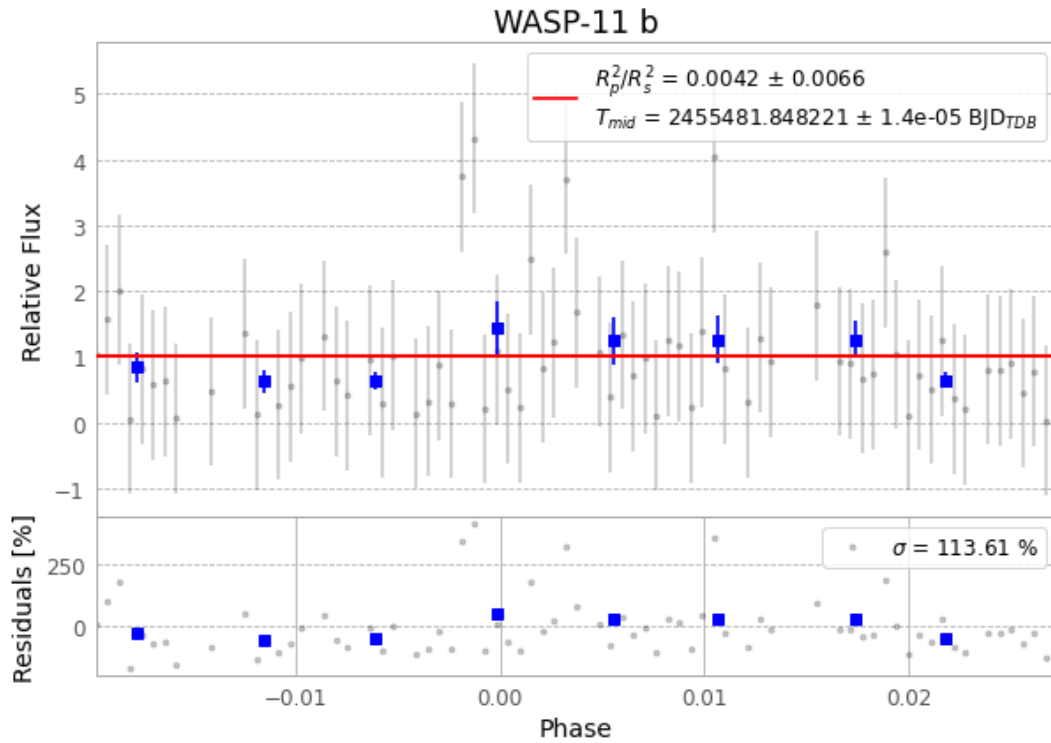


Figure 1 — FinalLightCurve_WASP-11 b_2010-10-11.png (SHA-256 c9ebfbad9269d625...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [406, 423], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7f9d9e63b5a77543...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

85 FITS frames (56 MB), HATP-10101012062714.FITS → HATP-10101012103914.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-101012.log (ce4737a6e1bf...), FinalLightCurve_WASP-11 b_2010-10-11.png (c9ebfbad9269...), FinalLightCurve_WASP-11 b_2010-10-11.csv (46f2da6b7262...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 23:33Z.